

Dr Waseem Raza, PhD

Machine Learning Engineer | AI/ML for Wireless & 5G/6G Systems

Wireless AI | 5G/6G | RAN Intelligence | Network Automation

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Professional Summary

Machine Learning Engineer and wireless-systems researcher with a PhD in Electrical and Computer Engineering and experience applying AI/ML to cellular-network diagnostics, propagation modeling, mobility, network resilience, and next-generation wireless systems. Industry and research work spans AT&T, AT&T Labs, Futurewei, and AI4Networks, with emphasis on reliable learning under domain shift, 5G/NR-NTN analysis, RAN optimization, outage management, digital-twin modeling, and intelligent network operations.

Technical Expertise, Leadership, and Professional Strengths

AI/ML for Wireless: Transfer learning, domain adaptation, adversarial learning, generative modeling, reinforcement learning, anomaly detection, classification, and robust model evaluation

Wireless Systems: LTE/4G, 5G-NR, NR-NTN, mobility, RAN, propagation, cell selection, user timing, network measurements, and service assurance

Network Intelligence: Root-cause analysis, outage detection and compensation, self-healing, optimization, resilience, and intelligent automation

Simulation & Digital Twins: 3GPP-compliant simulation, propagation modeling, synthetic data generation, digital twins, large-scale experimentation, and robustness testing

Programming & Data: Python, SQL, PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, visualization, and statistical analysis

Engineering & MLOps: Git, Docker, CI/CD, reproducible experimentation, APIs, cloud analytics, model validation, and production-oriented workflows

Leadership and Professional Strengths

- **Ownership and accountability:** Lead AI/ML initiatives through requirements analysis, data preparation, model development, validation, stakeholder review, and deployment planning.
- **Technical leadership:** Define technical approaches, evaluate alternatives, coordinate experimentation, and communicate findings to guide projects from ambiguous requirements to measurable outcomes.
- **Cross-functional collaboration:** Work with software engineers, network specialists, researchers, geologists, and public-safety stakeholders to translate domain requirements into practical data and AI solutions.
- **Continuous learning:** Apply ongoing study in generative AI, agentic AI, RAG, LLM evaluation, and cloud-native platforms through hands-on projects and professional coursework.
- **Reliability and commitment:** Maintain reproducible workflows, careful validation, and attention to detail throughout experimentation and delivery.
- **Adaptability and problem solving:** Combine critical thinking, systematic experimentation, and data-driven analysis to address unfamiliar technologies and complex operational problems.

Professional Experience

Machine Learning Engineer

AT&T, contractor via Innovatix Inc.

Feb. 2025-Present

Bedminster, NJ

- Own end-to-end development, testing, and operational integration of an AI-driven diagnostic system for critical E911 service-assurance workflows, improving automated root-cause classification accuracy from 20% to 80%.
- Designed domain-adaptive neural models and reusable data-processing workflows that generalize across heterogeneous operational domains and reduce performance degradation caused by distribution shifts.
- Developed multimodal AI prototypes combining work-order text, location information, network context, and operational signals to support GenAI-assisted diagnostics for VoNR and 5G service assurance.
- Created model-validation, error-analysis, and failure-mode assessment procedures to compare performance across domains and support reliable deployment in high-impact operational environments.
- Partnered with network engineers, software teams, and public-safety stakeholders to translate operational requirements into validated AI workflows and production-oriented data assets.

Advanced-Degrees Intern

AT&T Labs

Jun. 2024-Aug. 2024

Austin, TX

- Analyzed 500K+ LTE/5G-NR diagnostic logs and applied ML methods to detect anomalies, reducing diagnostic time by 20%.
- Investigated 5G NR-NTN cell-selection behavior and improved decision algorithms by 15% through data-driven analysis and validation.
- Conducted structured analysis of large operational datasets and communicated technical findings to engineering teams supporting next-generation wireless systems.

Advanced Wireless Technology Development Intern

Futurewei Technologies

Sep. 2022-Jan. 2023

Schaumburg, IL

- Improved autoencoder-based channel-state-information compression by 25% using transfer learning and systematic experimentation on large wireless datasets.
- Conducted 50+ experiments to validate model performance, robustness, and generalization under changing data conditions.
- Collaborated with research and engineering stakeholders to translate modeling results into recommendations for next-generation communication systems.

Advanced-Degrees Intern

AT&T Labs

- Designed ML and statistical methods for analyzing users' timing in 5G standalone networks, improving analytical precision by 20%.
- Analyzed large-scale user trace data with SQL and visualization tools and communicated findings to senior technical leadership.
- Converted raw operational traces into interpretable performance indicators to support diagnosis and network optimization.

Jun. 2022-Aug. 2022

Bedminster, NJ

Graduate Research Assistant

AI4Networks, University of Oklahoma

- Designed reusable Python simulation and data-generation platforms for large-scale experimentation, including a 3GPP-compliant network simulator and an AI-enabled 5G testbed.
- Built transfer-learning, adversarial-learning, GAN, autoencoder, and reinforcement-learning pipelines for sparse-data and cross-domain modeling.
- Developed robustness evaluation frameworks for models operating under distribution shifts, missing data, and limited labeled samples; led 100+ experiments from problem formulation through validation.
- Developed scalable self-healing and outage-management frameworks that improved modeled network resilience by 30%.
- Published peer-reviewed research, presented technical results internationally, and mentored junior researchers in ML implementation, experimental design, and technical writing.

Sep. 2019-Sep. 2024

Tulsa, OK

Education

PhD, Electrical and Computer Engineering

University of Oklahoma, Norman, Oklahoma, USA

09/2019-07/2025

MSc, Telecommunication Engineering

University of Engineering and Technology, Taxila, Pakistan

08/2014-08/2016

BSc, Telecommunication Engineering

University of Engineering and Technology, Taxila, Pakistan

11/2010-06/2014

Selected Projects

SyntheticNET: 3GPP-Compliant Network Simulation

Research Project

- Python-based simulation environment for realistic mobility, propagation, and data-driven experimentation in cellular networks.

Tools & methods: Python, 3GPP, Simulation, Wireless Networks

2020-2024

TurboRAN Experimental Platform

Research Infrastructure

- Multi-band, multi-tier experimental platform supporting evaluation of emerging AI-enabled radio-access-network solutions.

Tools & methods: 5G Testbeds, RAN, Experiment Design, Network Measurement

2020-2024

Certifications and Continuing Education

1. AI for Telecommunications Specialization — [AI CERTs / Coursera](#) | Completed (3/3) courses | June 2026 | [Certificate](#)

Focus: Three-course specialization connecting AI foundations with telecommunications network optimization, security, and customer experience.

Tools & Skills: AI for Telecommunications, Network Optimization, Cybersecurity, Customer Experience, Internet of Things, Python, TensorFlow

2. 4G Network Fundamentals — [Institut Mines-Télécom / Coursera](#) | Completed (1/1) courses | August 2026 | [Certificate](#)

Focus: Foundational study of fourth-generation cellular networks.

Tools & Skills: 4G Networks, Cellular Networks, Telecommunications

3. Business Considerations for 5G with Edge, IoT, and AI — [EDUCBA / Coursera](#) | Completed (1/1) courses | February 2026 | [Certificate](#)

Focus: Explores business considerations for combining 5G connectivity with edge computing, IoT, and AI.

Tools & Skills: 5G, Edge Computing, Internet of Things, Artificial Intelligence, Business Analysis

4. Machine Learning — Stanford University / Coursera | Completed

Focus: Core supervised and unsupervised learning, model evaluation, optimization, and practical ML system development.

Tools & Skills: Supervised Learning, Unsupervised Learning, Model Evaluation, Optimization

Selected Publications

On Multi-Parameter Optimization and Proactive Reliability in Cellular Networks

Journal article

2025

An AI-Driven Framework for Enhancing Resilience in Propagation Models to Enable Digital Twin

IEEE PIMRC

2024

Data-Driven Intelligent Outage Management for High Shadowing Environments in 5G&B Networks

IEEE SmartNets

2024

Toward a Hybrid RF/Optical Lunar Communication System (LunarComm)

IEEE Network

2022

